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PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR - CHENNAI



Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

AI-Based Electrical Fire Prevention System Using IOT Sensors and Machine Learning

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Introduction

Electrical fires are one of the leading causes of **property damage** and **loss of life**,often resulting from unnoticed **electrical faults**.Traditional fire alarm systems are mainly **reactive**,as they detect smoke or flames only **after a fire has already started**. This project aims to shift from **reactive** fire detection to **proactive fire prevention** by using real-time **sensor data** and **Machine Learning** to predict potential fire risks **before ignition occurs**. An **ESP32**-based IoT system continuously monitors important electrical and environmental parameters such as **temperature,voltage,current,and smoke** to identify **abnormal conditions**.Based on the collected data,the system automatically **classifies the level** of fire risk and takes preventive actions,including **power disconnection,buzzer activation,and SMS alerts**,thereby improving electrical safety and enabling early intervention.

ABSTRACT

Electrical fires are one of the leading causes of property damage and are often detected only after smoke or flames appear.This project proposes an **AI-Based Electrical Fire Prevention System** that predicts fire hazards before ignition using **IoT sensor and Machine Learning.**The system continuously monitors **temperature, voltage, current, and smoke levels** through an **ESP32 microcontroller**.A machine learning model analyzes the combined sensor data and classifies the fire risk into **Safe, Warning, High Risk, and Critical levels**.During critical conditions,the system automatically disconnects the power using a relay,activates a buzzer,sends an SMS alert through the **SIM800L GSM module**,and stores the event for future analysis.This low-cost,real-time solution improves electrical safety in homes,offices,laboratories,and industrial environments by **preventing fires** before they occur.

Literature Survey

AUTHOR NAME	FEATURES	METHODOLOGY	DATASET	ADDRESS	LIMITATIONS
Muhammed Fatih Peksen, Ulaş Yurtsever, Yılmaz Uyaroğlu (2024)	Temperature, humidity, CO, CO_2, voltage, current, cable temperature, outdoor temperature & humidity; IoT sensors; ML-based anomaly detection; pre-ignition prediction	Arduino Mega + Raspberry Pi 4; DHT22, MQ135, MQ4, PZEM-004T, MLX90640 sensors. ML models: DT, GNB, GPC, SVM with 5-fold cross-validation and RFE feature selection	3,478 samples collected experimentally. 3,025 normal (N) and 453 pre-ignition (51) samples. Features: temp., humidity, CO, CO_2, voltage, current, cable temp., outdoor temp. and humidity	Predicts electrical panel anomalies and early fire. risk for predictive maintenance using IoT and ML. GPC model achieved the best performance	Binary normal vs. pre-ignition classification due to limited S2-S4 samples. Dataset specific to the experimental panel; needs more diverse real-world electrical-fire data
Mengying Ma, Chengdi Xu, Junfeng Han (2025	Electrical parameter monitoring; arc detection; smart circuit breakers; EC-IoT cloud–edge–terminal architecture; BIM/digital building simulation; early-fire warning	EC-IoT cloud-edge-terminal architecture: Modbus edoe response model; electrical sensors, arc detection, smart circuit breakers: BIM/digital building simulation	No conventional public ML dataset. Uses electrical monitoring system data and digital building simulation	Provides early warning of electrical fires in high-rise residential. buildings and supports evacuation and fire-rescue decision making.	Focus mainly on electrical parameters; limited multi-source heterogeneous data integration; digital simulation accuracy can affect results
Weijie Xu, S. C. Chan, W. Y. Leong (2023)	AI-based fire-fighting supervision; facility monitoring; maintenance monitoring; staff training; questionnaire and field-investigation based evaluation	Mixed methods: questionnaires, interviews, field investigation, field visits, secondary data analysis and comparative case study	91 valid questionnaires from two public-building cases + interviews and field investigation secondary data	Evaluates effectiveness of AI in fire-fighting facility supervision, operation, maintenance and staff training	Small sample size and limited geographical coverage; case-geograph study based; not not nationally n representative
Abhishek Andil, Pranav Bhalerao, Gaurav Raygade, Saurav Bhosale, Pradnya Kasture (2025)	WSN; BLE beacons; ESP32/Raspberry Pi; fire-condition sensing; indoor location tracking; AI-based dynamic pathfinding; real-time fire detection and evacuation	WSN, BLE beacons, ESP32/Raspberry Pi, and AI-based dynamic pathfinding for real-time fire detection and evacuation routing	No conventional/public benchmark dataset reported. Uses sensor, fire-condition and indoor location information	Provides real-time indoor fire detection and personalized evacuation routes based on location and changing fire conditions.	Prototype level; limited real-world validation; high processing requirements; privacy and interoperability concerns.

Literature Survey

AUTHOR NAME	FEATURES	METHODOLOGY	DATASET	ADDRESS/ CONTRIBUTION	LIMITATIONS/ RESEARCH GAP
Jameson Mbale, Ndanji Simbeye (2025)	Temperature, gas concentration, smoke density; IoT-enabled sensors; automated alerts; fire suppression; risk classification (High/Low).	Arduino, IoT sensors, threshold logic, cloud alerts, automated suppression	240 simulated observations: temperature, gas, smoke	Low-cost real-time fire prevention and automated response; Zambia context	Simulated data; needs real-world pilot testing; power, false-positive and affordability issues
Abdullah Safi et al. (2022)	Wi-Fi + LoRaWAN; MQ2/gas sensor; flame sensor; environmental sensors; long-range communication; fault-tolerant fire monitoring.	Wi-Fi + LoRaWAN, Wi-Fi LoRa 32, gas/flame/environmental sensors	Experimental network data; 50 packets, 16–64 bytes, up to 600 m	Fault-tolerant long-range communication for smart-building fire monitoring	Mainly communication-focused; false alarms and network limitations
Aigulim Bayegizova et al. (2024)	YOLO/YOLOv4, CNN, Fast R-CNN; satellite imagery; FireWatch sensor images; deep-learning-based smoke/fire detection; wildfire monitoring	YOLOv4, CNN, Fast R-CNN	Satellite + FireWatch images; Kaggle/image datasets	Fast automated wildfire/fire detection; YOLOv4 reported 98.9% accuracy	Small/short fires difficult in satellite images; FireWatch coverage and cost limitations
Gwamzhi Ponsah Emmanuel et al. (2025)	CNN, YOLO, ResNet, MobileNet; multi-sensor data fusion; IoT; Kalman filtering; Bayesian fusion; knowledge distillation; edge/embedded systems.	CNN, YOLO, ResNet, MobileNet, IoT, multi-sensor fusion	FireNet, BoWFire, FASDD, D-Fire, PYRONEAR, and other publicly available fire/smoke datasets.	Combines multiple sensors and AI to improve robustness and reduce false alarms	High computation; limited standardization and real-world validation

Problem Statement

Electrical fires are mainly caused by **overheated wiring**, **voltage surges**, and **current overloads**. Traditional fire alarms detect **smoke** or **heat** only after a **fire starts**. An AI-based real-time system can predict **fire risks before ignition** and help **prevent accidents**.

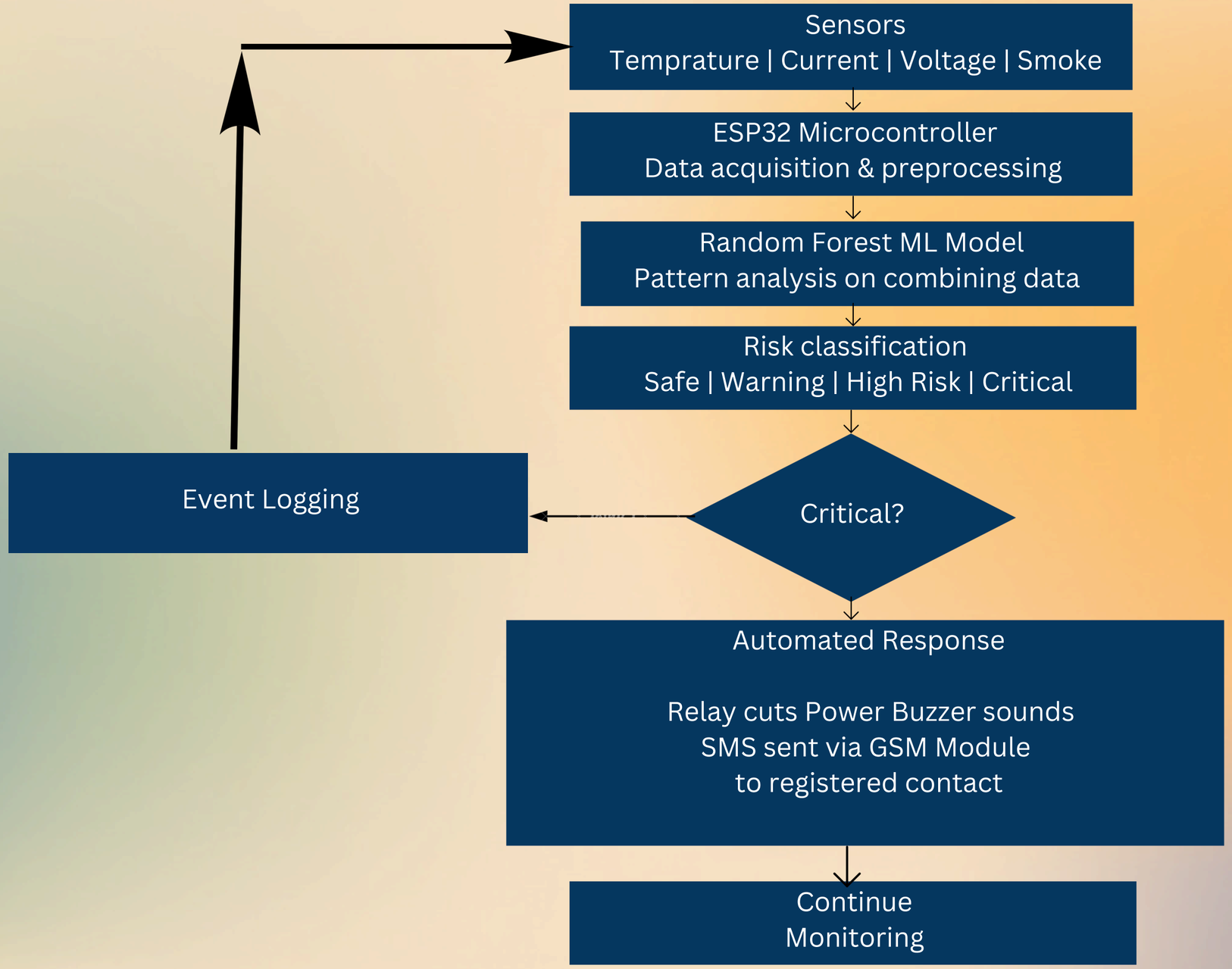
Objective

- To design a real-time **electrical fire monitoring system** using **ESP32** with **temperature,voltage,current,and smoke sensors.**
- To develop a **Random Forest Machine Learning model** that analyzes multi-sensor data to detect abnormal conditions that may lead to electrical fires.
- To classify fire risk into four levels: **Safe,Warning,High Risk,and Critical.**
- To automatically disconnect the power supply using a relay and activate a **buzzer** during critical conditions.
- To send **SMS alerts using the SIM800L GSM module** without requiring **Wi-Fi or Internet.**
- To log all sensor readings,risk levels,and system events for **monitoring,analysis,and future model improvement.**
- To develop a **cost-effective,scalable,and reliable** electrical fire prevention system for **homes,offices,laboratories,and industrial environments.**

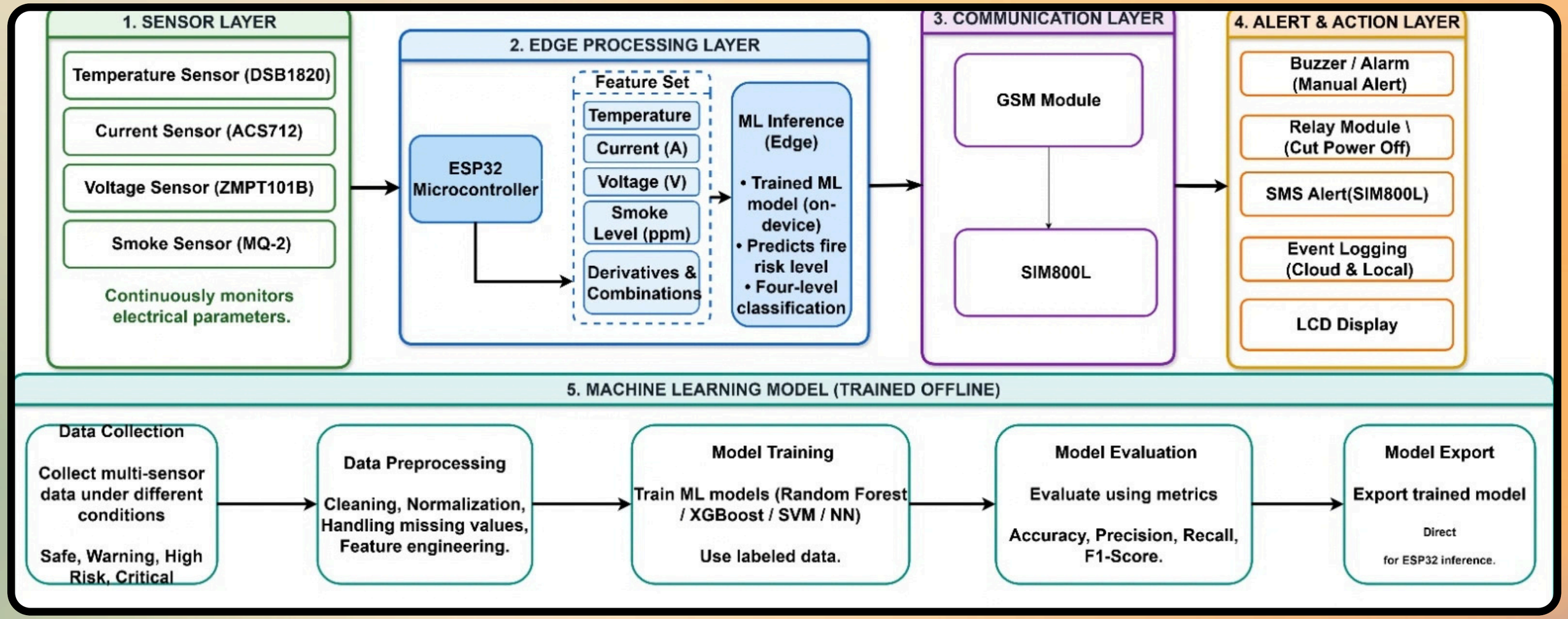
Methodology

The proposed system continuously collects **temperature,voltage, current, and smoke** readings using sensors connected to an **ESP32 microcontroller**.The collected sensor data is **filtered** and **preprocessed** to remove unwanted variations and prepare it for analysis. The preprocessed data is then given to the proposed **Random Forest machine learning classifier**,which analyses the sensor parameters and **predicts the electrical fire risk**.The predicted risk is classified into four levels: **Safe,Warning,High Risk, and Critical.** When a **Critical condition is detected,**the system automatically **disconnects the power supply** through a relay, activates a **buzzer,**and sends an SMS alert using the **SIM800L GSM module**.Finally,sensor readings, risk levels, and system actions are stored for monitoring, analysis, and future improvement of the model.

System Flow


Sensors
Temprature | Current | Voltage | Smoke
ESP32 Microcontroller
Data acquisition & preprocessing
Random Forest ML Model
Pattern analysis on combining data
Risk classification
Safe | Warning | High Risk | Critical
Critical?
Event Logging
Automated Response
Relay cuts Power Buzzer sounds
SMS sent via GSM Module
to registered contact
Continue
Monitoring

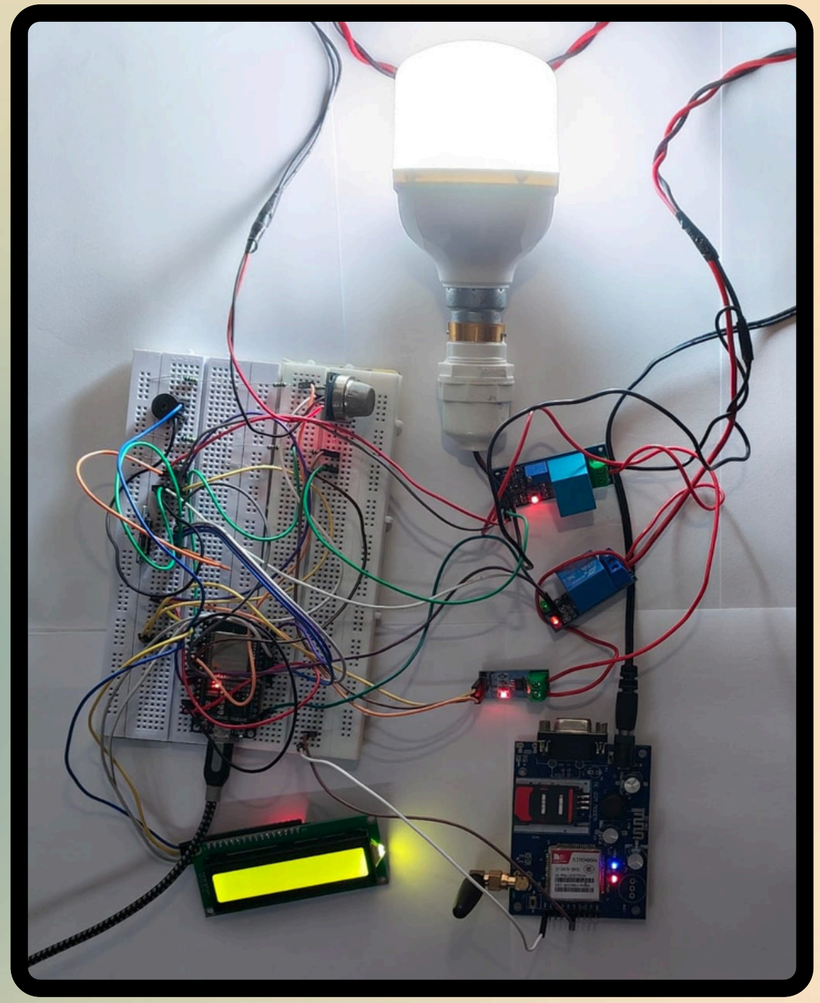
Architecture



Outcome

- A working prototype that utilizes **ESP32** to monitor **temperature,current,voltage,**and **smoke** in real-time.
- Risk levels were classified as **Safe,Warning,High Risk,and Critical based** on the trained **Random Forest ML model** and the **Decision Tree Model.**
- The system was able to **automatically cut off the power**,trigger a **buzzer**,and send an **SMS notification** upon detecting a Critical risk level.
- The prototype also had a **log of all the events** that occurred,including sensor reading,risk level,and action taken.
- A **low-cost and relliable system** that showcases the shift from traditional fire alarm systems to **AI-enabled** fire prevention systems.

AI-Based Fire Prevention System



1. **Power on:** The MQ-2 smoke sensor warms up first, because its readings aren't stable until then.

2. **Sensing:** The ESP32 reads four values: temperature, current, voltage and smoke.

3. **Combining:** it puts the four readings together as one input.

4. **Classifying:** the trained Decision Tree, running on the ESP32 itself, rates the situation as Safe, Warning, High Risk or Critical.
5. **Displaying:** the LCD shows the current risk level

6. **Acting on the level:**
- **Safe:** it keeps monitoring.
- **Warning:** it sends an SMS alert.
- **Critical:** the relay cuts the power to the load, the buzzer sounds, and an urgent SMS is sent.

7. **Alerting:** the SIM800L GSM module sends the SMS over the mobile network, so no Wi-Fi or internet is needed.

References

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Gantt Chart

ACTIVITES	WEEK 01	WEEK 02	WEEK 03	WEEK 04	WEEK 05	WEEK 06	WEEK 07	WEEK 08	WEEK 09	WEEK 10
Problem identification & Requirement Analysis	✓	✓								
Literature Survey		✓	✓							
System Design & Architecture			✓	✓						
Component selection & Procurement				✓	✓					
Hardware circuit Design					✓	✓				
Sensor integration & Data collection						✓	✓			
Data Preprocessing							✓			
ML Model Development & Risk Classification							✓			
Esp32 - ML Model Integration							✓			
Relay, Buzzer & GSM(SIM800L) Integration								✓		
Local Data Logging(MicroSD) & LCD Display								✓		
Testing & Performance Evaluation									✓	✓
Final Prototype Documentation & Presentation									✓	✓

Conclusion

In conclusion, we developed an AI-powered electrical fire prevention system using ESP32, IoT, and Machine Learning. The system continuously monitors electrical parameters to predict potential hazards at an early stage. During critical conditions, it automatically disconnects the power supply, activates an alarm, and sends SMS notifications. Overall, this system provides a smart, reliable, cost-effective, and scalable solution to improve electrical safety in various environments.

Thank You